

PRATYAKSH TOMAR

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TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript

Data Science & AI: NumPy, Pandas, Matplotlib, Scikit-learn, NLTK, VADER, Ollama, LangChain, RAG

Databases: MySQL, MongoDB, ChromaDB

Web Technologies: HTML5, CSS3, Tailwind CSS, React.js, Node.js, Express.js, MongoDB, Git/GitHub

Coursework: Data Structures, Algorithms, Object-Oriented Programming, Database Management System, Operating Systems, Computer Networks

PROJECTS

Resume Screening AI | *MERN Stack, Cohere AI, PDF-Parse, Firebase Auth, Multer*

[Live Link](#) | [Github](#)

- Built an **AI-driven** resume screening platform using the **MERN stack** that evaluates PDF resumes against job descriptions using Cohere's LLM API, returning automated match scores (**0–100%**) and actionable feedback.
- Engineered **secure backend services** in Express and MongoDB with **Multer** for file processing and Firebase Authentication using **High-Order Components (HOC)** to protect routes and support persistent user dashboards.
- Developed a full administrative suite and interactive frontend in React (Vite) featuring **Three.js** animations, personal history tracking, and cross-user submission management.

E-Consultation Sentiment Analysis Dashboard | *React.js, Python, Flask, NLTK, Chart.js*

[Live Link](#) | [Github](#)

- Architected an **end-to-end sentiment analysis engine** using **FastAPI** and **TextBlob** to process public feedback CSVs and categorize comments into positive, negative, and neutral sentiment categories.
- Implemented automated NLP pipelines incorporating **LSA-based extractive summarization (Sumy)** and **top-term frequency extraction** to turn large comment sets into concise executive summaries and word clouds.
- Designed an interactive data visualization interface using **Chart.js** to display dynamic KPIs and sentiment metrics, featuring automated PDF report generation (**via fpdf2**) complete with embedded analytics.

VectorDB | *C++, HNSW, KD-Tree, Ollama, REST API*

[Github](#)

- Engineered a high-performance C++ vector database supporting **HNSW**, **KD-Tree**, and **exact nearest-neighbor** search algorithms with cosine, Euclidean, and Manhattan distance metrics across 16D–768D spaces.
- Built a complete **Retrieval-Augmented Generation (RAG)** system integrating Ollama (nomic-embed-text and llama3.2) with custom 250-word sliding-window chunking for contextual document querying.
- Developed a **thread-safe C++ REST API** with **10+ endpoints** using `std::mutex` for concurrent access, backed by a dynamic 2D PCA scatter-plot UI for real-time vector clustering analysis.

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

Bachelor of Technology in Computer Engineering

2023 – Present

Dehradun Public School, Ghaziabad

Senior Secondary (Class XII) – CBSE

Percentage - **82.2%**

2020 – 2022

Dehradun Public School, Ghaziabad

Secondary (Class X) – CBSE

Percentage - **92%**

2018 – 2020

ACHIEVEMENTS

- Selected as a **finalist** in **Policy Conclave'26**, Indian Institute of Technology Kanpur.
- Participated in **Hacker's Premier League – Guru Gobind Singh Indraprastha University (Delhi)**.
- Competed in **National AI/ML Hackathon** by **AVEVA**, a national-level competition organised by IIT Hyderabad, building AI/ML solutions as part of team Thunders — Mar 2026.
- Awarded **Top Performer certificate** in **E&ICT IITG** Introduction to Programming by **Coding Ninjas × IIT Guwahati**, Feb 2025 (NSDC certified).
- Solved over **400+ coding problems** across platforms including **LeetCode** and **Code 360**.